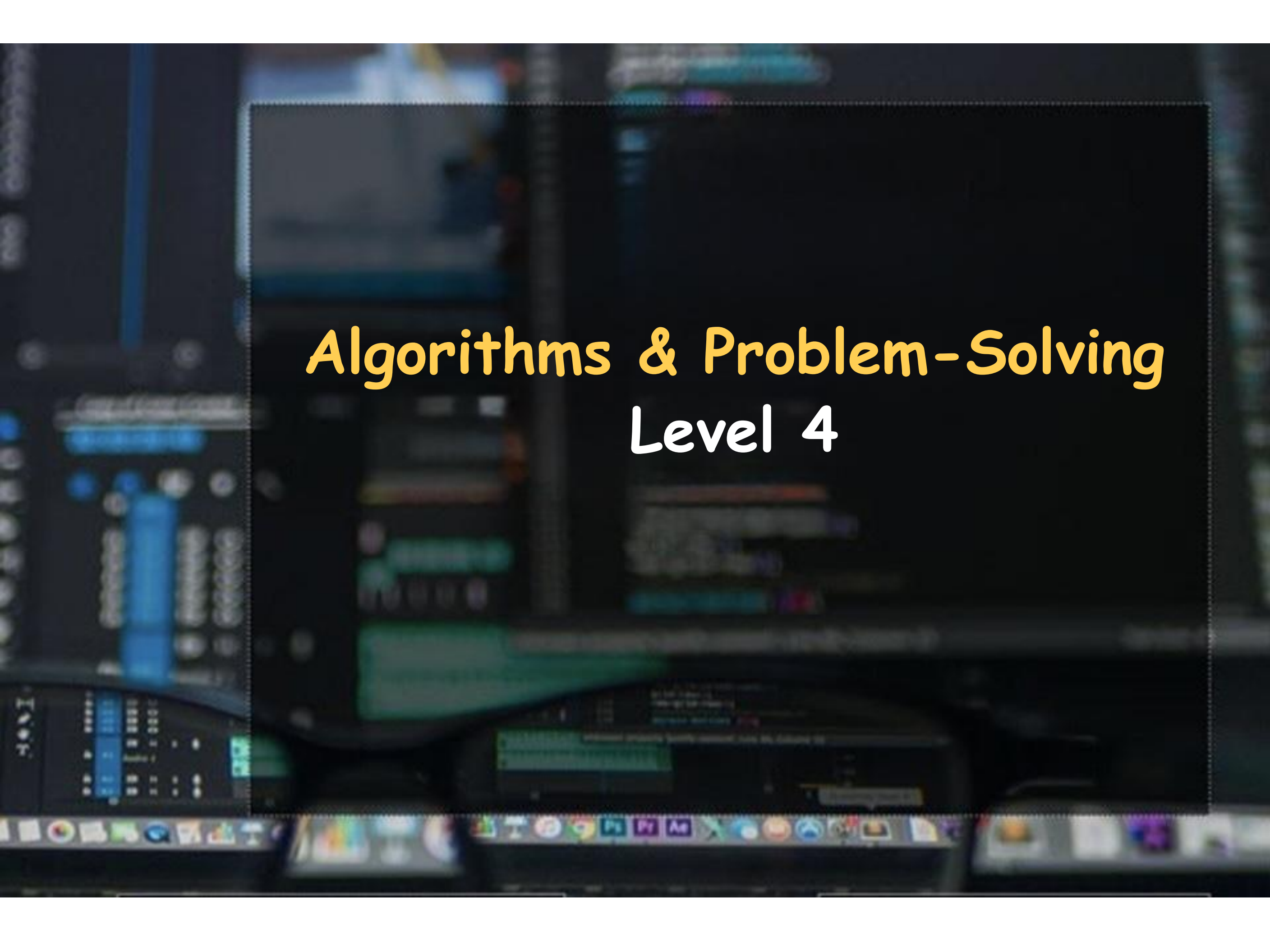




Algorithms & Problem-Solving Level 4

The background of the slide features a complex, abstract pattern of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, the silhouettes of several people are visible, appearing to be in a collaborative setting, possibly a meeting or a workshop, with some individuals leaning forward as if working on a table or screen.

Algorithms & Problem Solving

Level 4

Leap Year



Definition

- The task is to write a program in C++ that checks whether a given year is a leap year or not.
- A leap year is one that has 366 days instead of 365, and February has 29 days.



Explanation

- The program uses the following mathematical rule to determine a leap year:
 - A year is a leap year if: It is divisible exactly by 4, i.e., $\text{year} \% 4 == 0$
 - Except for years ending in "00" (centennial)
- These are only considered leap years if they are divisible exactly by 400, i.e., $\text{year} \% 400 == 0$ That is:
 - 2000 is a leap year because it is divisible by 400
 - 1900 is not a leap year because it is not divisible by 400, although it is divisible by 100
 - 1996 is a leap year because it is divisible only by 4

Procedures

① Read the year from the user:

 This is done using the `readYear()` function, which requests the year and returns it as an integer.

```
int readYear()
{
    int year; cout << "Please enter a year to check?";
    cin >> year; return year;
}
```

2 Checking if a year is a leap year:

■ This is done using the `isLeapYear(int year)` function, which applies the previous rules in order:

- If the year is divisible by 400, it is a leap year.
- Otherwise, if it is divisible by 100, it is not a leap year.
- Otherwise, if it is divisible by 4, it is a leap year.
- Otherwise, it is not a leap year.

```
bool isLeapYear(int year)
{
    if (year % 400 == 0)
        return true;
    else
    {
        if (year % 100 == 0)
            return false;
        else
            if (year % 4 == 0)
                return true;
            else
                return false;
    }
}
```

3 Displaying the result to the user:

■ In the `main()` function, the two previous functions are called, and then the appropriate result is printed:

```
if (isLeapYear(year))  
    cout << year << " is a leap year.";  
else  
    cout << year << " is not a leap year.";
```



Note:

- The program relies on recursion inside the `NumberToText()` function to break the number into parts (thousands, hundreds, tens, units).
- Attention should be paid to correct word forms (for example, "Hundred" instead of "Hundreds") to improve linguistic accuracy.



Example:

Year	Result
1996	Leap Year
1900	Not Leap Year
2000	Leap Year
2004	Leap Year
2010	Not Leap Year



Note:

- The division is checked using the remainder operator %
- The order of the conditions in the function is very important because the rule for 400 must be checked first



Explanation

- A year divisible by 100 is not always a leap year because the Gregorian calendar adds a day only every 4 years. However, to correct a slight discrepancy in the calculation, centuries not divisible by 400 are excluded from being leap years..

Summary

- ✓ A year is a leap year if it is divisible by 4 but not by 100, or if it is divisible by 400.
- ✓ The program reads the year from the user and prints the result directly.
- ✓ Some common examples:
 - 2000 is a leap year
 - 1900 is not a leap year
 - 2024 is a leap year

Thank you 😊